

CLUSTER STRUCTURES OF RICHARDSON VARIETIES

1. BRUHAT ATLAS

1.1. **Isomorphism.** For any $r \in W$, we have

$$\begin{aligned} (rU^-r^{-1} \cap U) \times (rU^-r^{-1} \cap U^-) &\xrightarrow{\sim} rU^-r^{-1}, & (g_1, g_2) &\mapsto g_1g_2 \\ (rU^-r^{-1} \cap U^-) \times (rU^-r^{-1} \cap U) &\xrightarrow{\sim} rU^-r^{-1}, & (h_1, h_2) &\mapsto h_1h_2. \end{aligned}$$

Define

$$\begin{aligned} \sigma_{r,-} : rU^-r^{-1} &\longrightarrow rU^-r^{-1} \cap U^-, & g_1g_2 &\mapsto g_2 \\ \sigma_{r,+} : rU^-r^{-1} &\longrightarrow rU^-r^{-1} \cap U, & h_1h_2 &\mapsto h_2. \end{aligned}$$

Consider

$$rU^-B/B \xrightarrow{\sim} BrB/B \times B^-rB/B, \quad grB \mapsto (\sigma_{r,+}(g)rB, \sigma_{r,-}(g)rB).$$

We have

$$\mathring{R}_{u,w} \cap rU^-B/B \xrightarrow{\sim} \mathring{R}_{u,r} \times \mathring{R}_{r,w}.$$

Actually,

$$\begin{aligned} gB \in B^-uB/B &\iff \sigma_{r,+}(g)B \in B^-uB/B, \\ gB \in BwB/B &\iff \sigma_{r,-}(g)B \in BwB/B. \end{aligned}$$

Assume $gB \mapsto (xB, yB)$, then xB is characterized to be the unique element of BrB/B of the form g_1gB for $g_1 \in rU^-r^{-1} \cap U^-$, and yB is characterized to be the unique element of B^-rB/B of the form g_2gB for $g_2 \in rU^-r^{-1} \cap U$. We view it as a rational function

$$\mathring{R}_{u,w} \dashrightarrow \mathring{R}_{u,r} \times \mathring{R}_{r,w}.$$

Example. Let us consider $r = s_2s_3s_2 \in S_4$. We have

$$rU^-r^{-1} = \left\{ r \begin{bmatrix} 1 & 0 & 0 & 0 \\ c & 1 & 0 & 0 \\ b & a & 1 & 0 \\ z & y & x & 1 \end{bmatrix} r^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ b & 1 & 0 & a \\ z & x & 1 & y \\ c & 0 & 0 & 1 \end{bmatrix} : x, y, z, a, b, c \in \mathbb{C} \right\}.$$

We have

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & a \\ & & 1 & y \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ b & 1 & & \\ z & x & 1 & \\ c & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ ac+b & 1 & 0 & a \\ yc+z & x & 1 & y \\ c & 0 & 0 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & & & \\ b & 1 & & \\ z & x & 1 & \\ c & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & a \\ & & 1 & y \\ & & & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ b & 1 & 0 & a \\ z & x & 1 & xa+y \\ c & 0 & 0 & 1 \end{bmatrix}.$$

As a result,

$$(\sigma_{r,+}, \sigma_{r,-}) : \begin{bmatrix} 1 & 0 & 0 & 0 \\ b & 1 & 0 & a \\ z & x & 1 & y \\ c & 0 & 0 & 1 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & a \\ & & 1 & y-xa \\ & & & 1 \end{bmatrix}, \begin{bmatrix} 1 & & & \\ b-ac & 1 & & \\ z-yc & x & 1 & \\ c & 0 & 0 & 1 \end{bmatrix}$$

1.2. Associativity. Note that this family of maps is not “associative”. That is, the following diagram in general does not commute

$$\begin{array}{ccc} \mathring{R}_{u,w} & \dashrightarrow & \mathring{R}_{u,r} \times \mathring{R}_{r,w} \\ \downarrow & \text{not commutative} & \downarrow \\ \mathring{R}_{u,s} \times \mathring{R}_{s,w} & \dashrightarrow & \mathring{R}_{u,r} \times \mathring{R}_{r,s} \times \mathring{R}_{s,w} \end{array}$$

Lemma. The diagram commute when r appears as a suffix of s .

Proof. Assume

$$\begin{array}{ccc} gB & \xrightarrow{\quad} & (agB, a'gB) \\ \downarrow & & \downarrow \\ (xgB, x'gB) & \xrightarrow{\quad} & (yxB, y'xB, x'gB) \end{array} \quad \begin{array}{ll} a \in rU^-r \cap U^- & a' \in rU^-r \cap U \\ b \in sU^-s \cap U^- & b' \in sU^-s \cap U \\ x \in sU^-s \cap U^- & x' \in sU^-s \cap U \\ y \in rU^-r \cap U^- & y' \in rU^-r \cap U \end{array}$$

By assumption $rU^-r \cap U \subseteq sU^-s \cap U$, $b'a' \in sU^-s \cap U$. Note that $x'gB$ is the unique element of $\mathring{R}_{s,w}$ of the form $x'gB$ for $x' \in sU^-s \cap U^-$. This proves $b'a'gB = x'gB$. Similarly $agB = yxB$. Next let us consider rational map

$$U^+ \times B^- \dashrightarrow B^- \times U^+, \quad (g_1, g_2) \mapsto (g'_1, g'_2)$$

such that $g_1g_2 = g'_1g'_2$. We have

$$\begin{array}{ccc} (rU^-r^{-1} \cap U) \times (rU^-r^{-1} \cap U^-) & \xrightarrow{\sim} & (rU^-r^{-1} \cap U^-) \times (rU^-r^{-1} \cap U) \\ \uparrow & & \uparrow \\ (rU^-r^{-1} \cap U) \times (sU^-s^{-1} \cap U^-) & \longrightarrow & (sU^-s^{-1} \cap U^-) \times (rU^-r^{-1} \cap U) \\ \downarrow & & \downarrow \\ (sU^-s^{-1} \cap U) \times (sU^-s^{-1} \cap U^-) & \xrightarrow{\sim} & (sU^-s^{-1} \cap U^-) \times (sU^-s^{-1} \cap U) \end{array}$$

Since the middle row is the intersection of upper and lower row, the middle row is an isomorphism. Assume $(y', x) \mapsto (\bar{b}, \bar{a}')$. Then $y'xgB = \bar{b}\bar{a}'gB$. Since $\bar{b} \in B^-$, we have $\bar{a}'gB = \bar{b}^{-1}y'xgB \in B^-rB/B$. This proves $\bar{a}'gB = a'gB$ by uniqueness. Since $y'xgB = \bar{b}\bar{a}'gB = \bar{b}a'gB$, we have $ba'gB = y'xB$ by uniqueness. $\square$

1.3. The function T . We know

$$\mathbb{C}^\times \xrightarrow{\sim} \mathring{R}_{r,rs_i}, \quad z \mapsto \dot{r}y_i(z)B.$$

Let $u \leq r < rs_i \leq w$. Let us define rational function

$$T_{r,rs_i} : \mathring{R}_{u,w} \dashrightarrow \mathring{R}_{r,rs_i} \xleftarrow{\sim} \mathbb{C}^\times.$$

Lemma. The rational map T_{r,rs_i} is given by

$$gB \mapsto \frac{\Delta_{\varpi_i}^{rs_i \varpi_i}(g)}{\Delta_{\varpi_i}^{r \varpi_i}(g)}.$$

In particular, when $u = r$, the function T_{r,rs_i} is regular; when $w = rs_i$, the function T_{r,rs_i}^{-1} is regular.

Proof. Let us prove it in three steps.

Step 1. It is true for $u = r < rs_i = w$.

Proof. We have

$$z = \Delta_{\varpi_i}^{s_i \varpi_i}(y_i(z)) = \frac{\Delta_{\varpi_i}^{s_i \varpi_i}(y_i(z))}{\Delta_{\varpi_i}^{\varpi_i}(y_i(z))} = \frac{\Delta_{\varpi_i}^{rs_i \varpi_i}(\dot{r}y_i(z))}{\Delta_{\varpi_i}^{r \varpi_i}(\dot{r}y_i(z))} = \frac{\Delta_{\varpi_i}^{rs_i \varpi_i}(g)}{\Delta_{\varpi_i}^{r \varpi_i}(g)}$$

for any representative g such that $gB = \dot{r}y_i(z)B$. $\square$

Step 2. It is true for $u = r < rs_i \leq w$.

Proof. The map $\mathring{R}_{r,w} \dashrightarrow \mathring{R}_{r,rs_i}$ is given by

$$gB \mapsto xgB, \quad \text{for some } x \in U^- \cap rs_i U^-(rs_i)^{-1}.$$

Since $U^- \cap rs_i U^-(rs_i)^{-1} \subset U^- \cap rU^-r^{-1}$, we have

$$x \in rs_i U^-(rs_i)^{-1}, \quad x \in rU^-r^{-1}.$$

This proves

$$\Delta_{\varpi_i}^{rs_i \varpi_i}(g) = \Delta_{\varpi_i}^{rs_i \varpi_i}(xg), \quad \Delta_{\varpi_i}^{r \varpi_i}(g) = \Delta_{\varpi_i}^{r \varpi_i}(xg).$$

Thus the formula is the same. $\square$

Step 3. It is true for $u \leq r < rs_i \leq w$.

Proof. The map $\mathring{R}_{u,w} \dashrightarrow \mathring{R}_{r,w}$ is given by

$$gB \mapsto xgB, \quad \text{for some } x \in U \cap rU^-r^{-1}.$$

Since $U \cap rU^-r^{-1} \subset U^- \cap rs_iU^-(rs_i)^{-1}$, we have

$$x \in rs_iU^-(rs_i)^{-1}, \quad x \in rU^-r^{-1}.$$

By similar reason as above, the formula is the same. □

We are done. □

For a sequence

$$e = u_0 < u_1 < \cdots < u_m = u \leq w$$

with $u_{j+1} = u_j s_{i_j}$ for some simple reflection s_{i_j} . Then we have birational map

$$\mathring{R}_{e,w} \dashrightarrow \mathring{R}_{u_0,u_1} \times \cdots \times \mathring{R}_{u_{m-1},u_m} \times \mathring{R}_{u,w} \cong (\mathbb{C}^\times)^m \times \mathring{R}_{u,w}$$

given by

$$gB \mapsto (T_{u_0,u_1}(gB), \dots, T_{u_{m-1},u_m}(gB), \dots)$$

Note that once setting $T_j = 1$ for $j < j_0$, then T_{j_0} is regular.

Theorem. We have

$$\mathbb{C}[\mathring{R}_{u,w}] = \mathbb{C}[\mathring{R}_{e,w}][T_{u_{i-1},u_i}^{\pm 1}] / \langle T_{u_{i-1},u_i} - 1 \rangle.$$

Proof. By induction, we need to show if $r < rs_i \leq w$, we have

$$\mathbb{C}[\mathring{R}_{rs_i,w}] = \mathbb{C}[\mathring{R}_{r,w}][T_{r,rs_i}^{\pm 1}] / \langle T_{r,rs_i} - 1 \rangle.$$

Note that T_{r,rs_i} is a regular function $\mathring{R}_{r,w} \rightarrow \mathbb{C}$. We reduce to show

$$\mathbb{C}[\mathring{R}_{r,w}][T_{r,rs_i}^{\pm 1}] \stackrel{(*)}{=} \mathbb{C}[\mathring{R}_{r,w} \cap rs_iU^-B/B] \cong \mathbb{C}[T_{r,rs_i}] \otimes \mathbb{C}[\mathring{R}_{rs_i,w}].$$

The open set rs_iU^-B/B is given by non-vanishing of minors

$$rs_iU^-B/B = \{gB \in G/B : \Delta_{\varpi_j}^{rs_i\varpi_j} \neq 0\}.$$

Note that if $j \neq i$, we already have $\Delta_{\varpi_j}^{rs_i\varpi_j}(g) = \Delta_{\varpi_j}^{r\varpi_j}(g) \neq 0$ for $gB \in \mathring{R}_{r,w} \subseteq B^-rB/B \subseteq rU^-B/B$. For $j = i$, we have $\Delta_{\varpi_i}^{rs_i\varpi_i}(g) = T_{r,rs_i}(g)\Delta_{\varpi_i}^{r\varpi_i}(g)$ for $gB \in \mathring{R}_{r,w} \subseteq B^-rB/B$. As a result,

$$\{gB \in \mathring{R}_{r,w} : T_{r,rs_i}(g) \neq 0\} = \mathring{R}_{r,w} \cap rs_iU^-B/B.$$

This proves (*). □

1.4. Constructing functions. If $u < s_i u \leq w < s_i w$, then there is a birational map

$$\mathring{R}_{u,s_i w} \dashrightarrow \mathring{R}_{u,s_i w}, \quad gB \longmapsto s_i gB.$$

Actually, $\overline{Bs_i w B/B}$ and $\overline{B^- u B/B}$ are both invariant under s_i .

Assume $s_i w > w$, consider

$$\begin{aligned} \pi_i^0 : \quad & \mathring{R}_{e,s_i w} \dashrightarrow \mathring{R}_{e,w} \times \mathring{R}_{w,s_i w} \xrightarrow{\text{proj}} \mathring{R}_{e,w} \\ \pi_i^1 : \quad & \mathring{R}_{e,s_i w} \xrightarrow{s_i} \mathring{R}_{e,s_i w} \dashrightarrow \mathring{R}_{e,w} \times \mathring{R}_{w,s_i w} \xrightarrow{\text{proj}} \mathring{R}_{e,w}. \end{aligned}$$

Lemma. Assume $r < rs_j < w < s_i w$ and $rs_j < s_i rs_j$, then

$$\pi_j^{0*}(T_{r,rs_j}) = T_{r,rs_j}, \quad \pi_j^{1*}(T_{r,rs_j}) = T_{s_i r, s_i rs_j}.$$

Proof. Note that $r < rs_j < s_i rs_j$ implies $r < s_i r$ for sake of length. Note that

$$Bs_i w B/B = Bs_i B \times_B BwB/B.$$

We can write a generic element of $Bs_i w B/B$ as $y_i(t)gB$ for $gB \in BwB/B$. Note that $y_i(t) \in U^- \cap wU^-w^{-1}$, as $w^{-1}\alpha_i > 0$. We have

$$\pi_j^0(y_i(t)gB) = gB.$$

By assumption, $s_j r^{-1}\alpha_i > 0$ and $r^{-1}\alpha_i > 0$,

$$y_i(t) \in rs_j U^-(rs_j)^{-1}, \quad y_i(t) \in rU^-r^{-1}.$$

We have

$$T_{r,rs_j}(y_i(t)gB) = \frac{\Delta_{\varpi_j}^{rs_j \varpi_j}(y_i(t)gB)}{\Delta_{\varpi_j}^{r \varpi_j}(y_i(t)gB)} = \frac{\Delta_{\varpi_j}^{rs_j \varpi_j}(gB)}{\Delta_{\varpi_j}^{r \varpi_j}(gB)} = T_{r,rs_j}(gB).$$

This proves the first formula.

Notice that

$$\Delta_{\lambda}^{r\lambda}(\ddot{s}_i g) = \Delta_{\lambda}^{s_i r \lambda}(g)$$

whenever $s_i r > r$. This proves the second formula. □

Let us fix a reduced word

$$w = s_{i_1} s_{i_2} \cdots s_{i_\ell}.$$

We can pick the left-most subword J for u . We introduce notations

$$w_{>k}^{\leq j} = s_{i_{k+1}} \cdots s_{i_j}, \quad u_{>k}^{\leq j} = s_{i_{k+1}}^{\delta_{k+1} \in J} \cdots s_{i_j}^{\delta_{k \in J}}.$$

By an abuse of notation, we also view them as the word.

Lemma. For any $k \leq j$, we have

$$u_{>k}^{\leq j} < s_{i_k} u_{>k}^{\leq j}.$$

Proof. Firstly it suffices to prove for $j = \ell$, since $u^{\leq j}$ is the also the left-most subword in $w^{\leq j}$. Assume $s_{i_k} u_{>k} < u_{>k}$, then s_{i_k} cancel with a simple reflection in $u_{>k}$, and so we can find another word left to $u_{>k}$. Formally, we have

$$u_{>k} = s_{i_k} \triangleright \cdots s_{i_1} \triangleright u = w_{\leq k}^{-1} \triangleright u$$

where the operator $\triangleright$ is characterized by $s_i \triangleright u = \min(u, s_i u)$. We automatically have $u_{>k} < s_{i_k} u_{>k}$ by definition. $\square$

Let us consider projections

$$\mathring{R}_{e,w} \xrightarrow{\pi_{i_1}^{\delta_1 \in J}} \mathring{R}_{e,w_{\geq 1}} \xrightarrow{\pi_{i_2}^{\delta_2 \in J}} \mathring{R}_{e,w_{\geq 2}} \dashrightarrow \cdots \dashrightarrow \mathring{R}_{e,w_{\geq \ell-1}} \xrightarrow{\pi_{i_\ell}^{\delta_\ell \in J}} \mathring{R}_{e,w_{\geq \ell}}.$$

For index $k \in J$ appears in u , i.e. i_k is chosen, we consider

$$T_{e,s_k} : \mathring{R}_{e,w_{\geq k}} \dashrightarrow \mathbb{C}^\times.$$

The pull-back of T_{e,s_k} along the chain will give $T_{u_{<k}, u_{\leq k}}$. The goal is to realize the projections as “cluster” morphism.

2. DOUBLE BS VARIETY

Let us define

$$\text{Conf}_w^u = \left\{ \begin{array}{ccc} y_0 U^- & \xrightarrow{v} & y_1 U^- \\ | & & | \\ x_0 U^+ & \xrightarrow{u} & x_1 U^+ \end{array} \right\} / G, \quad \begin{array}{ll} xU^+ \xrightarrow{u} yU^+ & \iff x^{-1}y \in U^+ \dot{u} U^+ \\ xU^- \xrightarrow{v} yU^- & \iff x^{-1}y \in U^- \ddot{v} U^- \\ xU^+ \text{ --- } yU^- & \iff x^{-1}y \in B^+ B^-. \end{array}$$

Theorem. We have

$$BuB \cap B^- v B^- \cong \text{Conf}_w^u.$$

Proof. The map is given by

$$g \mapsto \left[\begin{array}{ccc} U^- & \xrightarrow{v} & g_1 U^- \\ | & & | \\ t_2 U^+ & \xrightarrow{v} & g U^+ \end{array} \right] \quad \begin{array}{l} g = g_1 t_1 = t_2 g_2 \\ g_1 \in U^- \ddot{v} U^- \\ g_2 \in U \dot{u} U \\ t_1, t_2 \in T. \end{array}$$

The injectivity and surjectivity is not hard. $\square$

Let

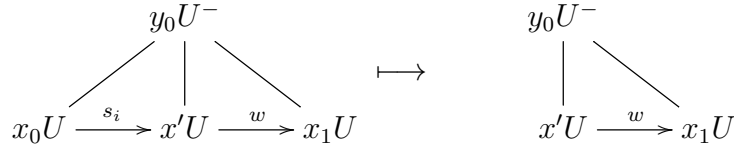
$$\begin{aligned} \tilde{R}_{u,w} &= \text{preimage of } \mathring{R}_{u,w} \text{ under } G/U \rightarrow G/B \\ &= BwB/U \cap B^- u B/U. \end{aligned}$$

We have $G_{e,w} \cong \tilde{R}_{e,w}$.

2.1. Operators. Let us describe some operators on configuration spaces. Assume $s_i w > w$.

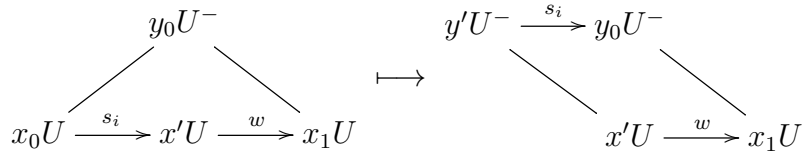
- We have a rational map

$$\text{Conf}_{s_i w} \dashrightarrow \text{Conf}_w .$$



- If $s_i w > w$, we have a regular map

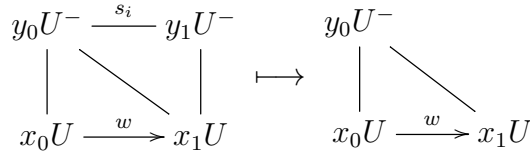
$$R_i : \text{Conf}_{s_i w} \longrightarrow \text{Conf}_w^{s_i} .$$



where $x_0^{-1} y' \in B s_i B^-$. This is known as the reflection morphism.

- We have a rational map

$$\text{Conf}_w^{s_i} \dashrightarrow \text{Conf}_w .$$



Now we have two rational maps

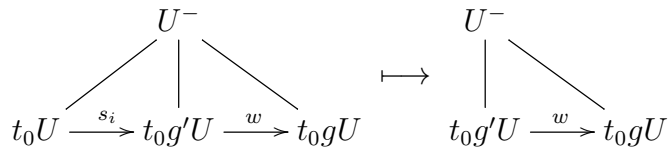
$$\text{Conf}_{s_i w} \dashrightarrow \text{Conf}_w, \quad \text{Conf}_{s_i w} \rightarrow \text{Conf}_w^{s_i} \dashrightarrow \text{Conf}_w .$$

Let us describe them.

Theorem. Both maps reduce to Richardson variety. We have

- (1) the first rational map is π_i^0 and
- (2) the second rational map is π_i^1 .

Proof. (1) The corresponding map is given by



We can assume $g' = y_i(t)$ for $t \in \mathbb{C}^\times$. Then

$$\begin{array}{ccc} U^- & & U^- \\ | & \searrow & | & \searrow \\ t_0 g' U & \xrightarrow{w} & gU & = & U & \xrightarrow{w} & t_0^{-1} y_i(-t) t_0 gU \end{array}$$

Note that $w^{-1}\alpha_i > 0$, we have

$$y_i(-t) \text{ and thus } t_0^{-1} y_i(-t) t_0 \in U^- \cap wU^- w^{-1}.$$

So the induced map is given by

$$t_0 g B \mapsto x \cdot t_0 g B$$

such that $x \in U^- \cap wU^- w^{-1}$ and $xgB \in BwB/B$.

(2) The reflection is given by

$$\begin{array}{ccc} \begin{array}{ccc} & U^- & \\ & | & \searrow \\ t_0 U & \xrightarrow{s_i} & t_0 g' U & \xrightarrow{w} & t_0 g U \end{array} & \mapsto & \begin{array}{ccc} \dot{s}_i U^- & \longrightarrow & U^- \\ & \searrow & \searrow \\ & t_0 g' B & \xrightarrow{w} & t_0 g B \end{array} \\ & & \\ \begin{array}{ccc} & \dot{s}_i U^- & \\ & | & \searrow \\ & t_0 g' U & \xrightarrow{w} & t_0 g U \end{array} & = & \begin{array}{ccc} & U^- & \\ & | & \searrow \\ t_0^{s_i} \cdot \dot{s}_i^{-1} g' U & \xrightarrow{w} & \dot{s}_i^{-1} t_0 g U \end{array} \end{array}$$

We can assume $g' = y_i(t)$ for $t \in \mathbb{C}^\times$, we have $\dot{s}_i^{-1} g' B = y_i(1/t)B$. By similar reason above, we are done. $\square$

2.2. Minors. For $xU^+ \longrightarrow yU^-$ and a dominant weight λ , we define

$$\Delta_\lambda(yU^-, xU^+) = \Delta_\lambda(xU^+, yU^-) = \Delta_\lambda(y^{-1}x).$$

Consider the following description of $\text{Conf}_w = \text{Conf}_w^e$.

$$\begin{array}{ccccccc} & & y_0 U^- & & & & \\ & \swarrow & & \searrow & & & \\ x_0 U & \xrightarrow{s_{i_1}} & x_1 U & \xrightarrow{s_{i_2}} & \cdots & \xrightarrow{s_{i_{\ell-1}}} & x_{\ell-1} U & \xrightarrow{s_{i_\ell}} & x_\ell U \end{array}$$

Adding a line gives an open condition. Let us add all lines to triangulate the diagram. By above, assume it is represented by $g \in BuB \cap B^-$ and we can fix $y_0 = 1$, $x_0 \in T$ and $x_\ell U = gU$. We can assume $x_\ell = z\dot{w}U$ for $z \in B$. Then we can write it as

$$\begin{array}{ccccccc} & & U^- & & & & \\ & \swarrow & & \searrow & & \searrow & \\ zU & \xrightarrow{s_{i_1}} & z\dot{s}_{i_1} U & \xrightarrow{s_{i_2}} & \cdots & \xrightarrow{s_{i_{\ell-1}}} & z\dot{w}_{\leq \ell-1} U & \xrightarrow{s_{i_\ell}} & z\dot{w}U \end{array}$$

Then we have

$$\Delta_\lambda(y_0 U^-, x_k U) = \Delta_\lambda^\lambda(z \dot{w}_{\leq k}) = \Delta_{w_{\leq k} \lambda}^\lambda(z).$$

After applying twist map, we get some Plücker coordinates.

Remark. If we want to get the classical minor, we shall use Conf_e^w , but instead of using the identification above, we use

$$\begin{array}{ccccccc} U^- & \xrightarrow{s_{i_1}} & \ddot{s}_{i_1} U^- & \xrightarrow{s_{i_2}} & \cdots & \xrightarrow{s_{i_{\ell-1}}} & \ddot{w}_{\leq \ell-1} U^- \xrightarrow{s_{i_\ell}} \ddot{w} U^- \\ & & \searrow & & & \nearrow & \\ & & & gU & & & \end{array}$$

Then we have

$$\Delta_\lambda(y_k U^-, x_0 U) = \Delta_\lambda^\lambda(\ddot{w}_{\leq k}^{-1} g) = \Delta_\lambda^{w_{\leq k} \lambda}(g).$$

Actually, the ring is

$$\mathbb{C}[B^- U / U \cap w B^- U / U]^{U^- \cap w U^- w^{-1}} \cong \mathbb{C}[\tilde{R}_{e,w}].$$

2.3. Cluster structure. The cluster structure of Conf_w^u can be described as follows. Let us pick a reduced word of (w, u) , viewed as an element of $W \times W$. The simple reflections are parametrized by

$$s_i = (s_i, e), \quad s_{\bar{i}} = (e, s_i).$$

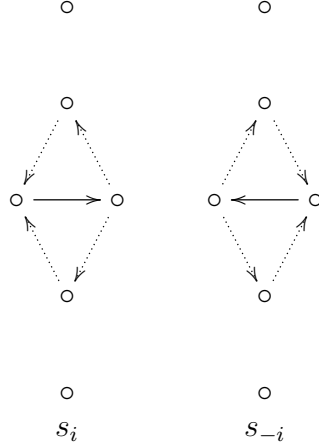
We define the quiver as follows. Consider $I \times \{0, \dots, \ell(u) + \ell(w)\}$. If the j -th word is s_i for $i \in I$,

$$\begin{array}{ll} \text{we glue for } i' \neq i & \text{we add arrows for } i' - i \\ (i', j-1) = (i', j); & (i, j) \dashrightarrow (i', j) = \\ \text{we add arrow} & (i', j-1) \dashrightarrow (i', j-1). \\ (i, j-1) \longrightarrow (i, j); & \end{array}$$

If the j -th word is $s_{\bar{i}}$ for $i \in I$,

$$\begin{array}{ll} \text{we glue for } i' \neq i & \text{we add arrows for } i' - i \\ (i', j-1) = (i', j); & (i, j) \dashleftarrow (i', j) = \\ \text{we add arrow} & (i', j-1) \dashleftarrow (i', j-1). \\ (i, j-1) \longleftarrow (i, j); & \end{array}$$

Here $--\rightarrow$ means $1/2$ many arrows. We frozen the vertices $I \times \{0, \ell(u) + \ell(w)\}$. Equivalently, it is obtained by gluing quivers



where arrow $\cdots\rightarrow$ means $1/2$ many arrows.

Each reduced word corresponding to a triangulation of the diagram

$$\begin{array}{ccccccc} y_0 U^- & \longrightarrow & y_1 U^- & \longrightarrow & \cdots & \longrightarrow & y_{\ell(u)} U^- \\ | & & & & & & | \\ x_0 U^+ & \longrightarrow & x_1 U^+ & \longrightarrow & \cdots & \longrightarrow & x_{\ell(w)} U^+ \end{array}$$

We define the cluster variable to be the minors. Let us call the leftmost edge by the 0-th edge. Then

$$\begin{array}{c} yU^- \\ \left| \begin{array}{c} j\text{-th edge} \\ xU \end{array} \right. \end{array} A_{(i,j)} = \Delta_{\varpi_i}(yU^-, xU^+) = \Delta_{\varpi_i}(y^{-1}x).$$

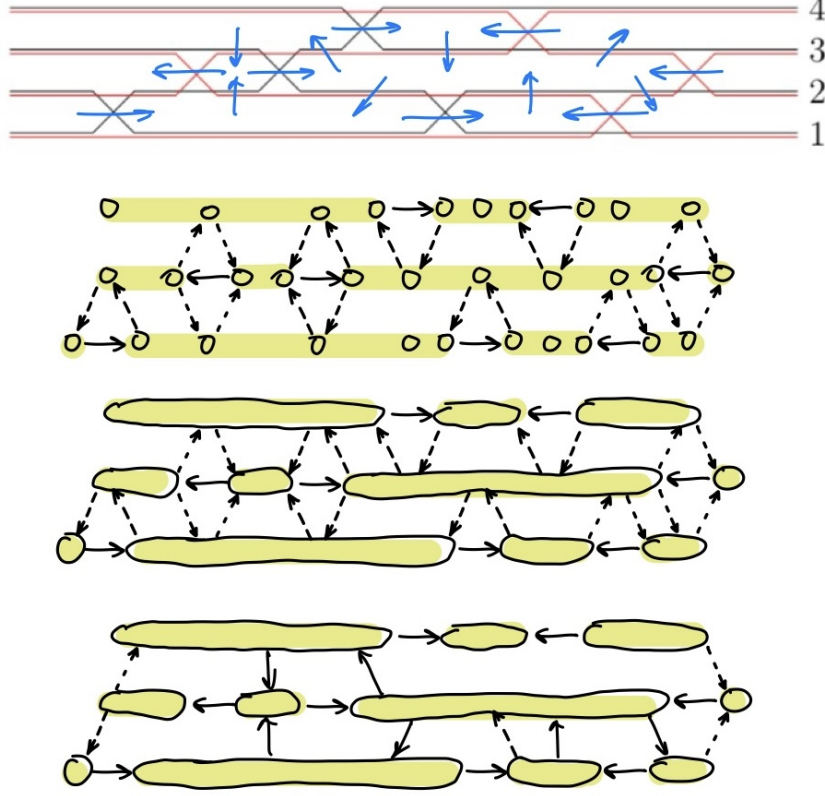
Different triangulation are related by mutation. That is,

$$\begin{array}{ccc} \begin{array}{|c|} \hline \xrightarrow{s_i} \\ \diagdown s_j \\ \hline \end{array} & \stackrel{i \neq j}{=} & \begin{array}{|c|} \hline \xrightarrow{s_i} \\ \diagup s_j \\ \hline \end{array} \end{array} \quad \begin{array}{ccc} \begin{array}{|c|} \hline \xrightarrow{s_i} \\ \diagdown s_j \\ \hline \end{array} & \stackrel{\mu(i,*)}{\longleftrightarrow} & \begin{array}{|c|} \hline \xrightarrow{s_i} \\ \diagup s_i \\ \hline \end{array} \end{array}$$

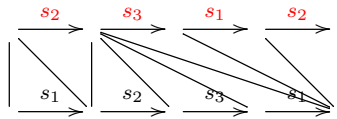
where the diagonal is the $*$ -th line.

In type A , this is equivalent to double wiring diagram.

Example. For example, in A_3 , we can construct using wiring diagram



It corresponds to the triangulation



It is direct to see if $(i, j - 1)$ and (i, j) are glued, then $A_{(i, j-1)} = A_{(i, j)}$.

3. RICHARDSON VARIETIES

3.1. Quasi cluster map. To prove this Theorem, we need the concept of quasi cluster map. Let

$$\epsilon_{ij} = \#\{i \rightarrow j\} - \#\{j \rightarrow i\}.$$

Let

$$p_{ij} = \begin{cases} \frac{1}{2}c_{i'j'}, & i = (i', 0), j = (j', 0) \text{ or } i = (i', \ell), j = (j', \ell). \\ 0, & \text{otherwise.} \end{cases}$$

Note that p_{ij} is not anti-symmetric. We define the X -variable to be

$$X_j = \prod_i A_i^{\epsilon_{ij} + p_{ij}}.$$

There is closed formula for mutation of X -variables. For a unfrozen vertex k

$$\mu_k(X_i) = \begin{cases} X_k^{-1}, & i = k, \\ X_i(1 + X_k^{-1})^{-m}, & i \neq k, \epsilon_{ik} = m \geq 0, \\ X_i(1 + X_k)^m, & i \neq k, \epsilon_{ik} = -m \leq 0. \end{cases}$$

In particular, for $i \notin \{k_1, \dots, k_m\}$, e.g. when i is frozen,

$$(\mu_{k_m} \circ \dots \circ \mu_{k_1})(X_i) = X_i \cdot (\text{a function in } X_{k_1}, \dots, X_{k_m}).$$

The function only depends on the full sub quiver generated by $\{i, k_1, \dots, k_m\}$.

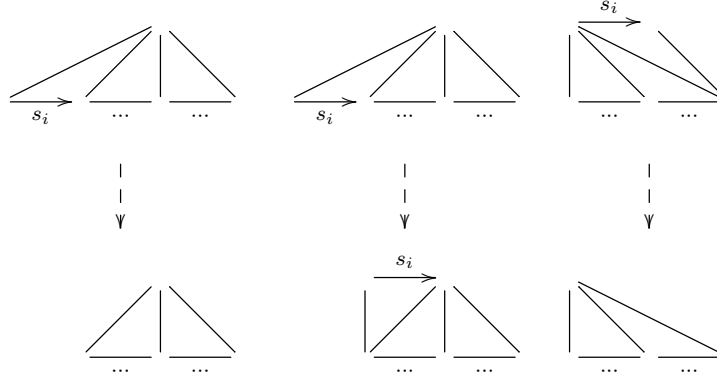
Definition. A quasi-cluster map is an inclusion of fractional fields of cluster algebras preserving unfrozen X -variables. That is, we have two injections

$$\sigma : \text{Frac}(\mathcal{A}_Q) \rightarrow \text{Frac}(\mathcal{A}_{Q'}), \quad \iota : Q_0^{\text{uf}} \rightarrow Q_0^{\text{uf}}$$

with $\sigma(X_i) = X_{\iota(i)}$ for unfrozen $i \in Q_0^{\text{uf}}$.

Lemma. Quasi-cluster map is preserved under mutation.

Theorem. The three operators



Proof. This follows from direct computation. □

Moreover, we can describe the rightmost frozen variables

- We can extend the quiver embedding such that it includes the right-most frozen vertices. This extension exists for any direction of mutation.

- The maps

$$\text{Conf}_{s_i w} \dashrightarrow \text{Conf}_w, \quad R_i : \text{Conf}_{s_i w} \dashrightarrow \text{Conf}_w^{s_i}$$

also preserves the rightmost X -variables, i.e.

$$X_{j, \ell(w)} \mapsto X_{j, \ell(s_i w)}, \quad X_{j, 1 + \ell(w)} \mapsto X_{j, \ell(s_i w)}$$

- The morphism

$$\text{Conf}_w^{s_i} \dashrightarrow \text{Conf}_w$$

satisfies

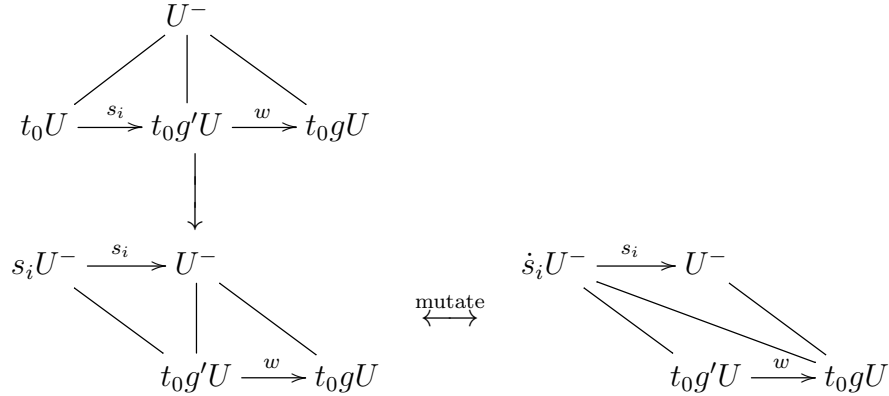
$$X_{(j, \ell(w))} \mapsto \begin{cases} X_{(i, \ell(w))} A_{(i, 1 + \ell(w))}, & j = i, \\ X_{(j, \ell(w))} A_{(i, \ell(w))}^{-a_{ij}} A_{(i, 1 + \ell(w))}^{a_{ij}}, & j \neq i. \end{cases}$$

Here $(i, 1 + \ell(w))$ is the rightmost i -vertex in $\text{Conf}_w^{s_i}$, $(i, 1 + \ell(w))$ is the image of the rightmost i -vertex in Conf_w .

3.2. The initial step. Recall the map

$$R_i : \text{Conf}_{s_i w} \longrightarrow \text{Conf}_w^{s_i}.$$

We have



The mutation μ is the composition of mutations mutable vertices of level i from left to right. As a result,

$$T_{e, s_i}(t_0 g B) = \frac{\Delta_{\varpi_i}^{s_i \varpi_i}(t_0 g)}{\Delta_{\varpi_i}^{s_i \varpi_i}(t_0 g)} = R_i^* \left(\frac{\Delta_{\varpi_i}(\dot{s}_i U^-, t_0 g U)}{\Delta_{\varpi_i}(U^-, t_0 g U)} \right)$$

That is

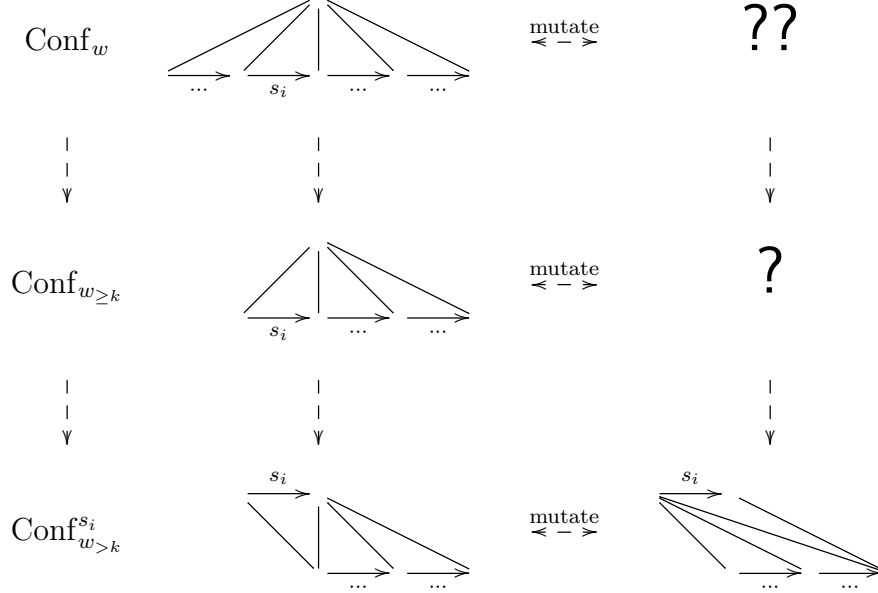
$$T_{e, s_i} = R_i^* \left(\frac{A_{(i, \ell(w))}}{A_{(i, \ell(w) + 1)}} \right),$$

where $\{A_*\}$ is the cluster variable of the southeast corner.

Let $u = s_i$ is a simple reflection, appeared as the k -th simple reflection in w . Let us consider

$$\Phi : \text{Conf}_w \dashrightarrow \text{Conf}_{w_{\geq 1}} \dashrightarrow \cdots \dashrightarrow \text{Conf}_{w_{\geq k}} \dashrightarrow \text{Conf}_{w_{> k}}^{s_i}$$

The following is the diagram



The mutation is given by the obvious inclusion map of the unfrozen part of the quiver. By above,

$$T_{e, s_i} = \Phi^* \left(\frac{A_{(i, \ell(w) - k)}}{A_{(i, \ell(w) - k + 1)}} \right)$$

where $\{A_*\}$ is the cluster variable of the southeast corner.

Theorem. We have

$$T_{e, s_i} = \frac{X_{(i, \ell(w)) + 1}}{A_{(i, \ell(w) + 1)}} = \frac{A_{(i, \ell(w))}}{A_{(i, \ell(w) + 1)}} \cdot \left(\begin{array}{l} \text{a Laurent monomial in } A_* \\ \text{for } * \text{ connected to } (i, \ell(w) + 1) \end{array} \right),$$

where $\{A_*\}$ is the cluster variable at $??$. Note that in our convention, there could be arrows between frozen vertices.

Proof. The proof uses the concepts introduced below. To clarify, let us name the cluster variables, pulled back by the downward arrows.

$$\begin{array}{ccc} \{\bar{A}\} \& \{\bar{X}\} & \xleftarrow{\text{mutate}} \xrightarrow{\text{mutate}} \{A\} \& \{X\} \\ \downarrow & & \downarrow \\ \{\bar{A}'\} \& \{\bar{X}'\} & \xleftarrow{\text{mutate}} \xrightarrow{\text{mutate}} \{A'\} \& \{X'\} \end{array}$$

There is an index shift. To eliminate this ambiguity, we name the different vertices of i -level from right to left by $(i, -1)$, $(i, -2)$ etc.

We have

$$T_{e,s_i} = \frac{A'_{(i,-2)}}{A'_{(i,-1)}} = \frac{X'_{(i,-1)}}{A'_{(i,-1)}}.$$

Since $(i, -1)$ is frozen, we have

$$A'_{(i,-1)} = \bar{A}'_{(i,-1)} = \bar{A}_{(i,-1)} = A_{(i,-1)}.$$

We have

$$\begin{aligned} X'_{(i,-1)} &= \bar{X}'_{(i,-1)} \cdot (\text{a function in unfrozen } \bar{X}_*) \\ &= \bar{X}_{(i,-1)} \cdot (\text{a function in unfrozen } \bar{X}_*) \\ X_{(i,-1)} &= \bar{X}_{(i,-1)} \cdot (\text{the same function in unfrozen } \bar{X}_*). \end{aligned}$$

To see the last equality, it suffices to notice that the quiver embedding can be extended to $(i, -1)$. As a result,

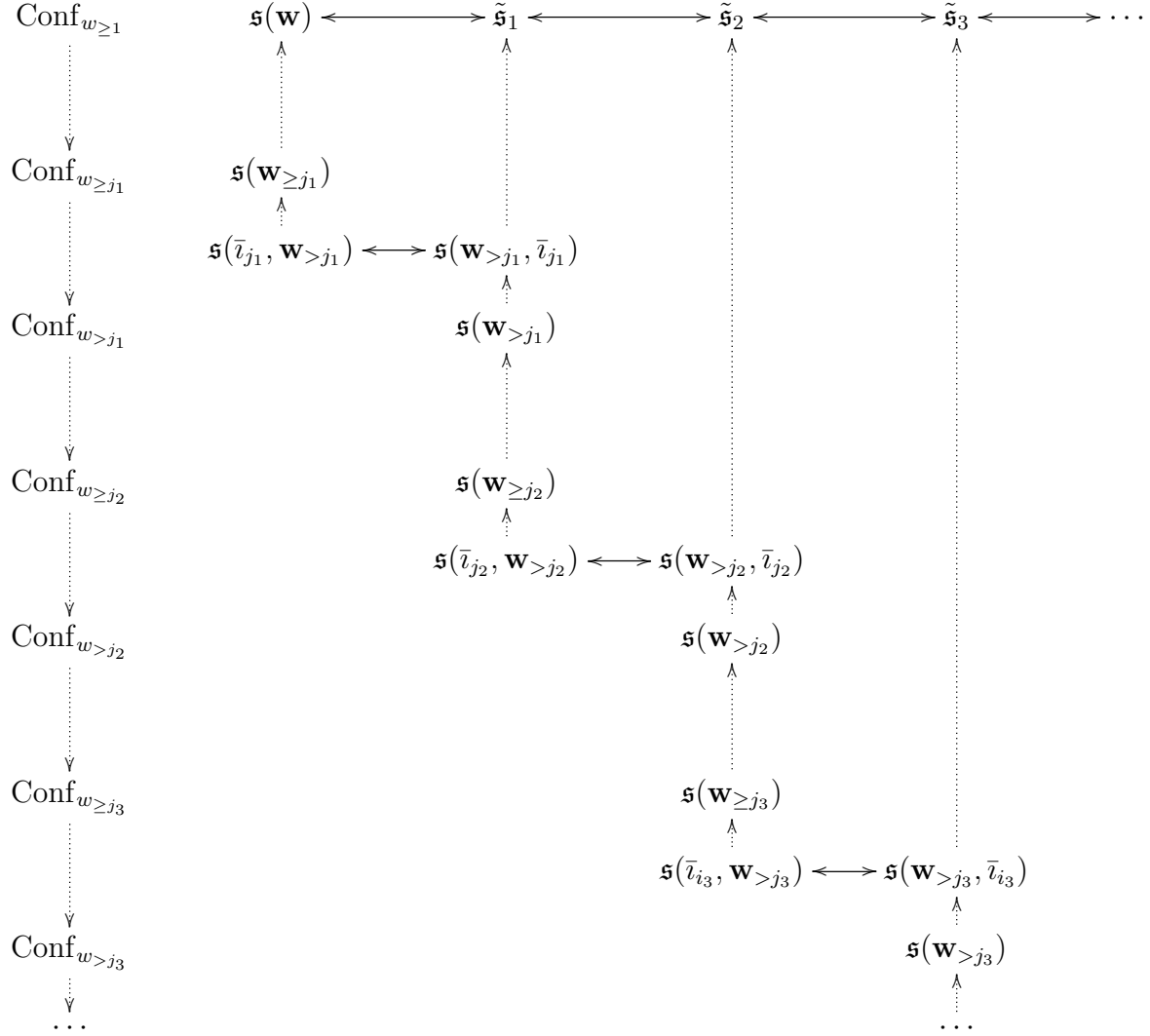
$$T_{e,s_i} = \frac{X_{(i,-1)}}{A_{(i,-1)}} = \frac{A_{(i,-2)}}{A_{(i,-1)}} \left(\begin{array}{l} \text{a Laurent monomial in } A_* \\ \text{for } * \text{ connected to } (i, -1) \end{array} \right).$$

Here we use the quiver embedding to conclude that there is an arrow $(i, -2) \longrightarrow (i, -1)$. This is exactly what we need. $\square$

3.3. Seeds. For a reduced word $\mathbf{w}$, we denote $\mathfrak{s}(\mathbf{w})$ the corresponding seed for double BS variety. Recall

$$\mathfrak{s}(s_{\bar{i}} \oplus \mathbf{w}) \xleftarrow{\mu} \mathfrak{s}(\mathbf{w} \oplus s_{\bar{i}})$$

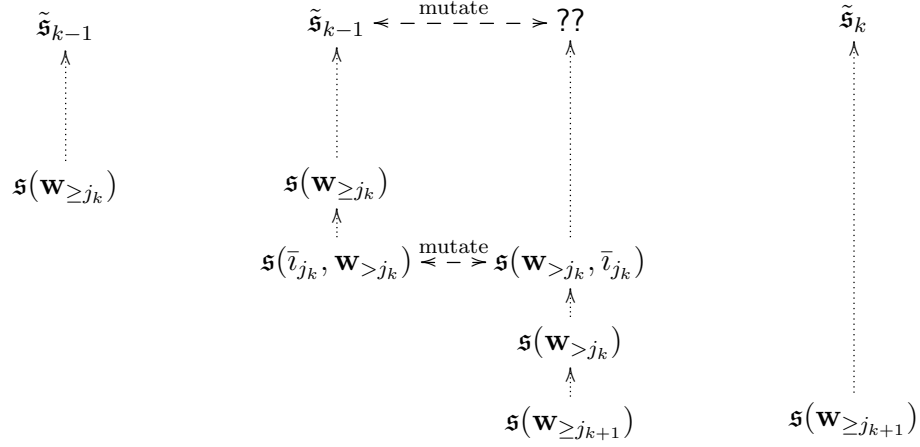
where μ is the sequence of mutations of unfrozen i -vertices from left to right. Let us consider



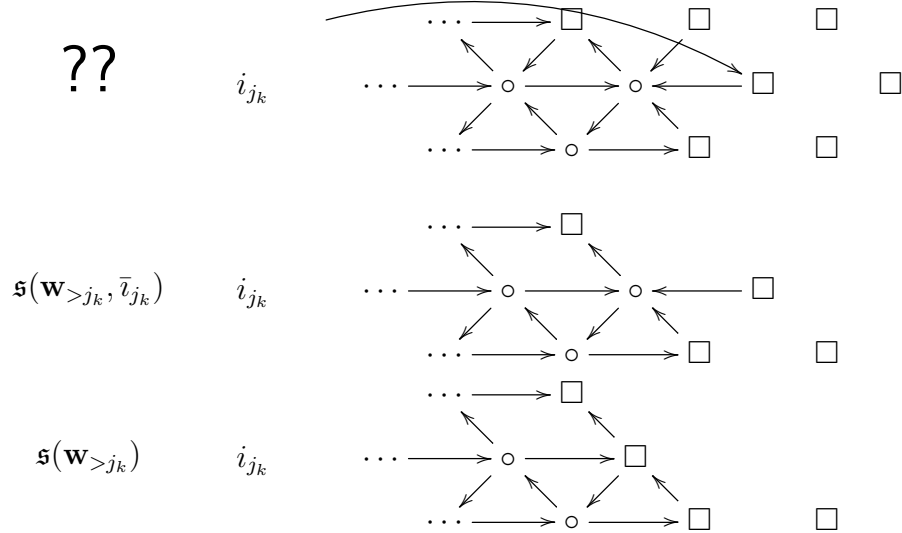
Here we take $\tilde{s}_0 = s(w)$. Assume we are given

$$s(w_{\geq j_k}) \dashrightarrow \tilde{s}_{k-1}$$

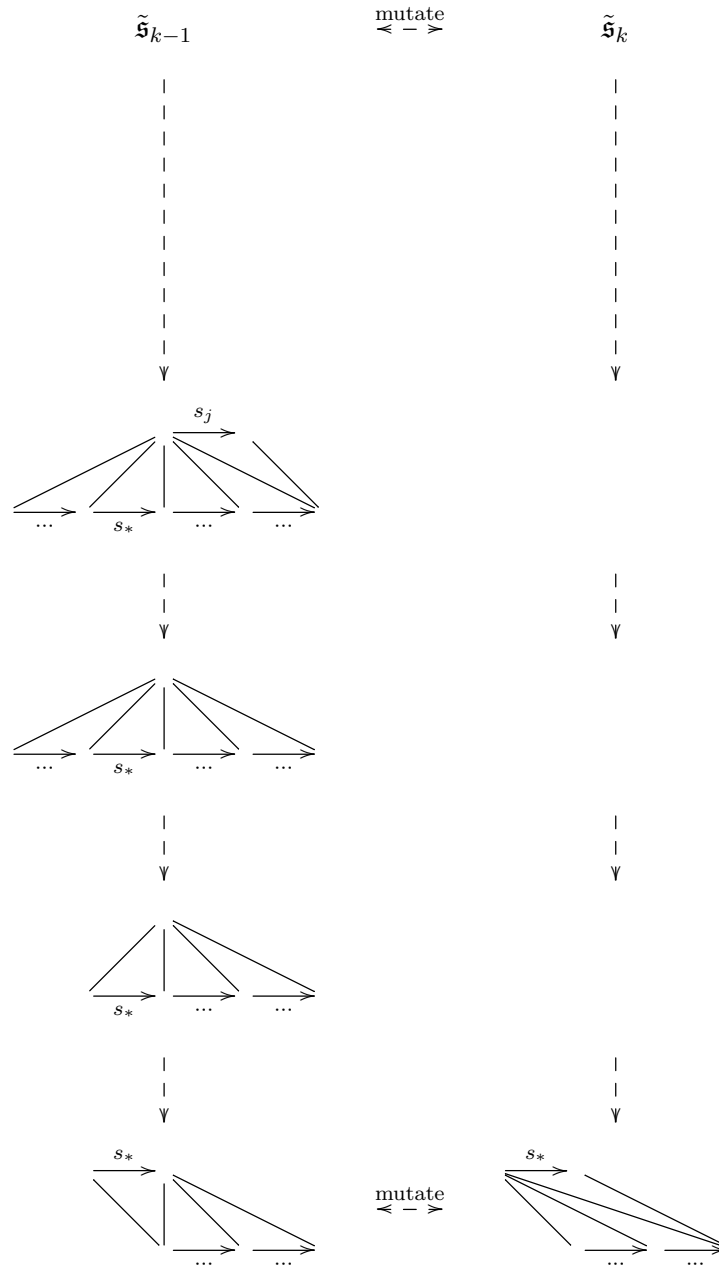
a quiver embedding with unfrozen vertices sent to unfrozen vertices. Let us construct



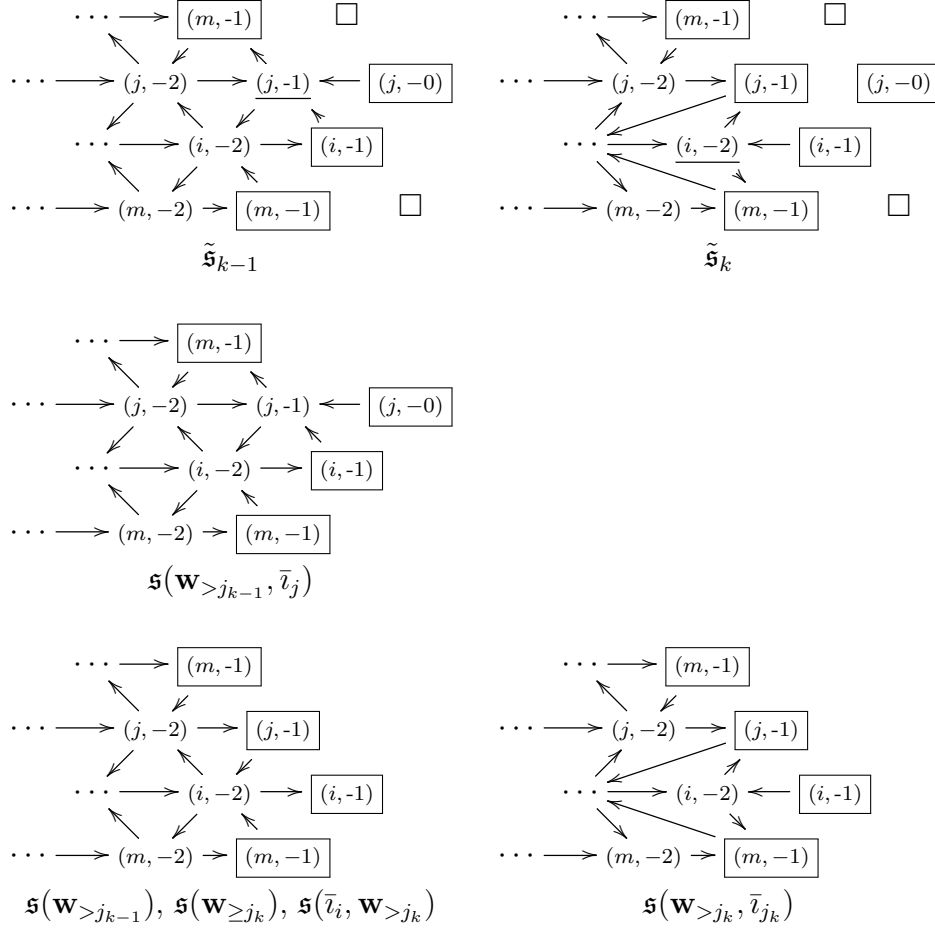
The mutation is composition of mutation of unfrozen vertices. Here $\tilde{\mathfrak{s}}_k$ is obtained from ?? by freezing vertices connected to the rightmost i_{j_k} -vertex of $\mathfrak{s}(\mathbf{w}_{> j_k}, \bar{i}_{j_k})$. Note that these vertices cannot be the image of unfrozen vertices of $\mathfrak{s}(\mathbf{w}_{\geq j_{k+1}})$. That is,



3.4. **General Cases.** To proceed induction, we need to consider



Let $j = j_{k-1}$ and $i = j_k$, and name the vertices as follows.



Let us name the cluster variables, pulled back by the downward arrows

$$\begin{aligned}
 (\dagger) \quad & \{\bar{A}\} \ \& \ \{\bar{X}\} \xleftrightarrow{\text{mutate}} \{A\} \ \& \ \{X\} \\
 (*) \quad & \{\bar{A}''\} \ \& \ \{\bar{X}''\} \\
 (**) \quad & \{\bar{A}'\} \ \& \ \{\bar{X}'\} \xleftrightarrow{\text{mutate}} \{A'\} \ \& \ \{X'\}
 \end{aligned}$$

$\uparrow$ $\uparrow$
 $\vdots$ $\vdots$

Theorem. For any $m \in I$, we have

$$A'_{(m, -1)} \sim A_{(m, -1)}, \quad X'_{(m, -1)} \sim X_{(m, -1)}$$

where $f \sim g$ if

$$f = g \cdot \left(\begin{array}{l} \text{a Laurent monomial in } A_* \\ \text{for frozen } * \neq (m, -1) \text{ in } \mu(\tilde{\mathfrak{s}}_{k-1}) \end{array} \right), \quad \mu = \text{mutate}.$$

Proof. It is obvious that $\sim$ does not depend on the choice of seed, as it involves only frozen variables. Moreover if we write $\sim = \sim_k$ to emphasize the dependence of k , then $f \sim_{k-1} g \Rightarrow f \sim_k g$ since we have more frozen variables in $\mu(\tilde{\mathfrak{s}}_{k-1})$ and when $(m, -1)$ is not the old $(m, -1)$, it is unfrozen.

For any $m \in I$, since mutations do not involve any $(m, -1)$, we have

$$A'_{(m,-1)} = \bar{A}'_{(m,-1)} = \bar{A}''_{(m,-1)}.$$

When $m \neq j$, since $(m, -1)$ is rightmost, we have

$$\bar{A}''_{(m,-1)} \sim \bar{A}_{(m,-1)} = A_{(m,-1)} \quad (\text{by induction}).$$

When $m = j$,

$$\begin{aligned} A'_{(j,-1)} &= \bar{A}''_{(j,-1)} \\ &= \bar{X}''_{(j,-0)} \sim \bar{X}_{(j,-0)} \quad (\text{by induction}) \\ &= X_{(j,-0)} \sim A_{(j,-0)}. \quad (\text{by definition}) \end{aligned}$$

Here we used the fact that $(j, -0)$ does not connected any $(i, *)$ in $\mathfrak{s}(\mathbf{w}_{>j_{k-1}}, \bar{l}_j)$.

When $m = j$, we have

$$\begin{aligned} X'_{(j,-1)} &= \bar{X}'_{(j,-1)} \cdot (\text{a function in unfrozen } \bar{X}'_*) \\ &= \bar{X}''_{(j,-1)} \cdot \bar{A}''_{(j,-0)} \cdot (\text{a function in unfrozen } \bar{X}''_*) \\ &\sim \bar{X}''_{(j,-1)} \cdot (\text{a function in unfrozen } \bar{X}''_*) \\ &= \bar{X}_{(j,-1)} \cdot (\text{a function in unfrozen } \bar{X}_*) = X_{(j,-1)}. \end{aligned}$$

When $m \neq j$,

$$\begin{aligned} X'_{(m,-1)} &= \bar{X}'_{(m,-1)} \cdot (\text{a function in unfrozen } \bar{X}'_*) \\ &= \bar{X}''_{(m,-1)} \cdot (\text{a function in unfrozen } \bar{X}''_*) \cdot \bar{A}''^{-a_{jm}}_{(i,-1)} \bar{A}''^{a_{jm}}_{(i,-0)} \\ &= \bar{X}_{(m,-1)} \cdot (\text{a function in unfrozen } \bar{X}_*) \cdot \bar{X}''^{-a_{jm}}_{(j,-0)} \bar{A}''^{a_{jm}}_{(j,-0)} \\ &= X_{(m,-1)} \cdot \bar{X}''^{-a_{jm}}_{(j,-0)} \bar{A}''^{a_{jm}}_{(j,-0)} \\ &\sim X_{(m,-1)} \cdot \bar{X}''^{-a_{jm}}_{(j,-0)}. \end{aligned}$$

Note that

$$\bar{X}^{-a_{jm}}_{(j,-0)} = \left(\begin{array}{c} \text{a Laurent monomial in } \bar{A}_* \\ \text{for } * \text{ connected to } (j, -0) \end{array} \right) \sim 1.$$

Note that by definition, these $*$'s are frozen, and in particular do not change under μ . $\square$

From the construction, for each $j_k \in J$, there is an $f(j_k) = (i_{j_k}, *)$ corresponding to the rightmost i_j -vertex of $\mathfrak{s}(w_{>j_k, \bar{l}_{j_k}})$, i.e. we need to freeze vertices connected to $f(j_k)$ to obtain $\tilde{\mathfrak{s}}_k$. Note that $f(j_k)$ itself is frozen in $\tilde{\mathfrak{s}}_{k-1}$.

Theorem. For $j_0 \in J$, we have

$$T_{u < j_0, u \leq j_0} = \frac{X_{f(j_0)}}{A_{f(j_0)}} \cdot \left(\begin{array}{c} \text{a Laurent monomial of} \\ \text{frozen variables} \neq A_{f(j)} \text{ for } j \geq j_0 \end{array} \right).$$

Proof. The initial case is proved before. The induction follows directly from above. Note that $A_{(m,-1)}$ and $X_{(m,-1)}$ are frozen in $\tilde{\mathfrak{s}}_k$, so will not be changed later. $\square$

Let $\tilde{\mathfrak{s}}$ be the seed obtained finally. Let $\mathfrak{s}$ be the seed obtained from $\tilde{\mathfrak{s}}$ by deleting vertices $f(j)$ for $j \in J$ and all leftmost i -vertices.

Theorem. We have

$$\mathbb{C}[\tilde{R}_{u,w}][T_{u < j, u \leq j}^{\pm 1} : j \in J] = \text{the upper cluster algebra of } \tilde{\mathfrak{s}}.$$

Proof. Since once we invert $T_{u < j, u \leq j}$ for all $j < j_0$, the variable $A_{f(j_0)}$ and the monomial above are already inverted. So it suffices to invert $X_{f(j_0)}$, i.e. freeze the vertices connected to $f(j_0)$. $\square$

4. EXAMPLE

In the example, I will use

$$\Delta_A^B(g) = \det(g_{ij})_{i \in B, j \in A} = \Delta_{\epsilon_A}^{\epsilon_B}(g).$$

Note that A is for columns, and B is for rows.

Let us consider $W = S_5$, i.e.

$$\begin{aligned} w &= s_1 s_3 s_4 s_2 s_3 s_1 s_2 s_4 s_1, \\ u &= s_1 \cancel{s_3} \cancel{s_4} s_2 s_3 s_1 \cancel{s_2} \cancel{s_4} \cancel{s_1}. \end{aligned}$$

We will represent

$$\begin{array}{ccc} & U^- & \\ & \swarrow \quad \searrow & \\ zU^+ & \xrightarrow[w]{} & zwU^+ \end{array} \quad z \in B^+ \cap wB^-w^{-1}.$$

We have

$$B \cap wB^-w^{-1} = \left\{ \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ 0 & x_6 & x_7 & x_8 & x_9 \\ 0 & 0 & x_{10} & x_{11} & x_{12} \\ 0 & 0 & 0 & x_{13} & x_{14} \\ 0 & 0 & 0 & 0 & x_{15} \end{bmatrix} : \begin{array}{l} x_7 = 0 \\ x_1 x_6 x_{10} x_{13} x_{15} = 1 \end{array} \right\}.$$

Let

$$z = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ 0 & x_6 & 0 & x_8 & x_9 \\ 0 & 0 & x_{10} & x_{11} & x_{12} \\ 0 & 0 & 0 & x_{13} & x_{14} \\ 0 & 0 & 0 & 0 & x_{15} \end{bmatrix}, \quad z\dot{w} = \begin{bmatrix} x_5 & -x_4 & x_2 & x_3 & x_1 \\ x_9 & -x_8 & x_6 & 0 & 0 \\ x_{12} & -x_{11} & 0 & x_{10} & 0 \\ x_{14} & -x_{13} & 0 & 0 & 0 \\ x_{15} & 0 & 0 & 0 & 0 \end{bmatrix}$$

We have

$$\begin{aligned} T_{e, s_1} &= \frac{\Delta_1^2(z\dot{w})}{\Delta_1^1(z\dot{w})} = \frac{x_9}{x_5} \\ T_{s_1, s_1 s_2} &= \frac{\Delta_{12}^{23}(z\dot{w})}{\Delta_{12}^{12}(z\dot{w})} = \frac{-x_9 x_{11} + x_8 x_{12}}{-x_5 x_8 + x_4 x_9} \\ T_{s_1 s_2, s_1 s_2 s_3} &= \frac{\Delta_{123}^{234}(z\dot{w})}{\Delta_{123}^{123}(z\dot{w})} = \frac{-x_6 x_{12} x_{13} + x_6 x_{11} x_{14}}{x_5 x_6 x_{11} - x_2 x_9 x_{11} - x_4 x_6 x_{12} + x_2 x_8 x_{12}} \\ T_{s_1 s_2 s_3, s_1 s_2 s_3 s_1} &= \frac{\Delta_1^3(z\dot{w})}{\Delta_1^2(z\dot{w})} = \frac{x_{12}}{x_9}. \end{aligned}$$

The seed is presented in Figure 1. If $|A| = k$, then

$$\text{the cluster variable of vertex } A = \Delta_A^{[k]}(z).$$

For example, in $\tilde{\mathfrak{s}}_0$, we have

$$\begin{aligned} A_{\boxed{1}} &= \Delta_1^1(z) = x_1 & A_{\boxed{123}} &= \Delta_{123}^{123}(z) = x_1 x_6 x_{10} \\ A_2 &= \Delta_2^1(z) = x_2 & A_{124} &= \Delta_{124}^{123}(z) = x_1 x_6 x_{11} \\ A_4 &= \Delta_4^1(z) = x_4 & A_{\boxed{245}} &= \Delta_{245}^{123}(z) = x_5 x_6 x_{11} - x_2 x_9 x_{11} - x_4 x_6 x_{12} + x_2 x_8 x_{12} \\ A_{\boxed{5}} &= \Delta_5^1(z) = x_5 & A_{\boxed{1234}} &= \Delta_{1234}^{1234}(z) = x_1 x_6 x_{10} x_{13} \\ A_{\boxed{12}} &= \Delta_{12}^{12}(z) = x_1 x_6 & A_{1245} &= \Delta_{1245}^{1234}(z) = -x_1 x_6 x_{12} x_{13} + x_1 x_6 x_{11} x_{14} \\ A_{24} &= \Delta_{24}^{12}(z) = -x_4 x_6 + x_2 x_8 & A_{\boxed{2345}} &= \Delta_{2345}^{1234}(z) = -x_5 x_6 x_{10} x_{13} + x_2 x_9 x_{10} x_{13} + x_3 x_6 x_{12} x_{13} \\ A_{\boxed{45}} &= \Delta_{45}^{12}(z) = -x_5 x_8 + x_4 x_9 & & + x_4 x_6 x_{10} x_{14} - x_2 x_8 x_{10} x_{14} - x_3 x_6 x_{11} x_{14} \end{aligned}$$

In other diagrams, we also use

$$\text{the cluster variable of vertex } A^B = \Delta_A^B(z).$$

By computation, we have

$$\begin{aligned} A_? &= x_1 x_8 \\ A_{??} &= x_1 x_9 \\ A_{???} &= x_1 x_8 x_{12} - x_1 x_9 x_{11} \\ A_{????} &= x_1 x_6 x_{12}. \end{aligned}$$

Note that $f(j)$ and $X_{f(j)}/A_{f(j)}$ are

$$\begin{aligned}\frac{X_{\boxed{5}}}{A_{\boxed{5}}} &= \frac{A_{??}}{A_{\boxed{5}}} = \frac{x_1 x_9}{x_5} = A_{\boxed{1}} T_{e, s_1} \\ \frac{X_{\boxed{45}}}{A_{\boxed{45}}} &= \frac{A_{???}}{A_{\boxed{45}}} = \frac{x_1 x_8 x_{12} - x_1 x_8 x_{11}}{x_4 x_9 - x_5 x_8} = A_{\boxed{1}} T_{s_1, s_1 s_2} \\ \frac{X_{\boxed{245}}}{A_{\boxed{245}}} &= \frac{A_{1245} A_{???}}{A_{\boxed{245}}} = \frac{-x_1 x_6 x_{12} x_{13} + x_1 x_6 x_{11} x_{14}}{x_5 x_6 x_{11} - x_2 x_9 x_{11} - x_4 x_6 x_{12} + x_2 x_8 x_{12}} A_{\boxed{???}} = A_{\boxed{1}} A_{\boxed{???}} T_{s_1 s_2, s_1 s_2 s_3} \\ \frac{X_{\boxed{??}}}{A_{\boxed{??}}} &= \frac{A_{????}}{A_{\boxed{??}}} = \frac{x_1 x_6 x_{12}}{x_1 x_9} = A_{\boxed{12}} A_{\boxed{1}}^{-1} T_{s_1 s_2 s_3, s_1 s_2 s_3 s_1}.\end{aligned}$$

After inverting $T_{*,*}$,

$$A_{??}, A_{???}, A_{1245}, A_{????}$$

all become invertible, and they are frozen in the $\tilde{\mathfrak{s}}_4$. To set $T_{*,*} = 1$, we need to set

$$\begin{aligned}A_{\boxed{5}} &= A_{??} A_{\boxed{1}}^{-1} \\ A_{\boxed{45}} &= A_{???} A_{\boxed{1}}^{-1} \\ A_{\boxed{245}} &= A_{1245} A_{\boxed{???}}^{-1} A_{\boxed{1}}^{-1} \\ A_{\boxed{??}} &= A_{????} A_{\boxed{12}}^{-1} A_{\boxed{1}}\end{aligned}$$

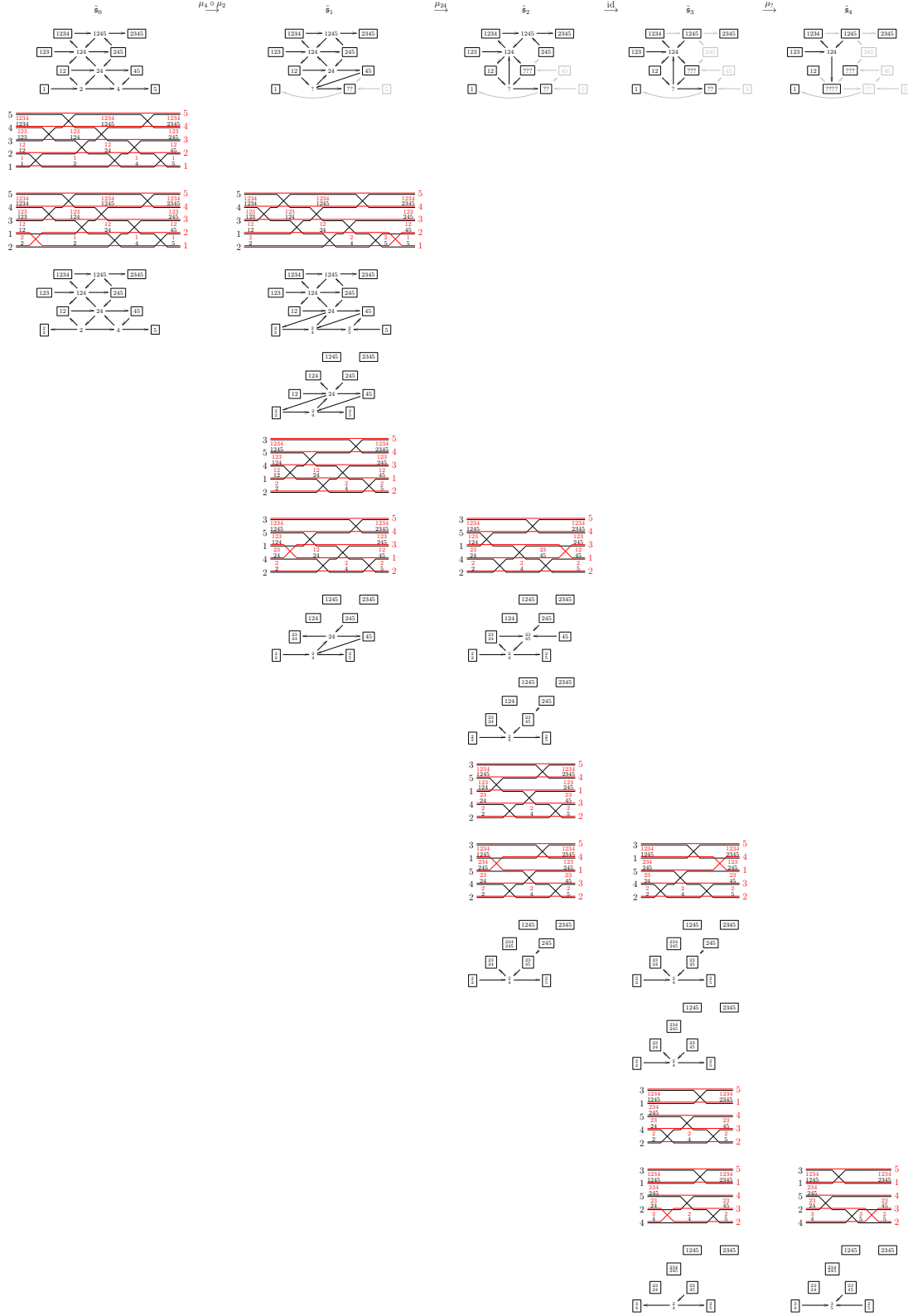


FIGURE 1. The seed